Factor Analysis 39

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Table of Contents

Overview

- *Factor analysis* (FA) is developed in psychological study and widely used in behavioral and social science.
- In a survey with 100 questions about human behaviours, the essential traits are energy focus(extraversion/introversion), decision making(thinking/feeling), etc.
- In a financial market with 500 stocks, key factors such as global economic conditions and country-specific policies play crucial roles in influencing stock prices.
- The essential purpose is to describe the covariance relationships among many variables in terms of a potentially smaller number underlying latent random quantities called *factors*.

FA vs. PCA

- Both PCA and FA can be viewed as attempts to approximate the covariance matrix. The primary question in factor analysis is whether the data are consistent with a prescribed structure.
- PCA transforms observed variables to obtain principal components, while FA transforms factors to obtain observed variables.

Table of Contents

The factor model

Let $\mathbf{X}$ be a p-dim observable random vector with $\boldsymbol{\mu} = E(\mathbf{X})$.
The factor model with m common factors is

$$\mathbf{X}_{p\times 1} = \boldsymbol{\mu}_{p\times 1} + \mathbf{L}_{p\times m}\mathbf{F}_{m\times 1} + \boldsymbol{\epsilon}_{p\times 1}$$

and in the component form:

here $F_1, F_2, \cdots, F_m$ is scale.

1-st factor

$$\begin{aligned} X_1 &= \mu_1 + \ell_{11}F_1 + \ell_{12}F_2 + \ldots + \ell_{1m}F_m + \epsilon_1 \\ X_2 &= \mu_2 + \ell_{21}F_1 + \ell_{22}F_2 + \ldots + \ell_{2m}F_m + \epsilon_2 \\ &\vdots \\ X_p &= \mu_p + \ell_{p1}F_1 + \ell_{p2}F_2 + \ldots + \ell_{pm}F_m + \epsilon_p. \end{aligned}$$

weight in p-dimensions

where l_{ij} is called the *loading* of the i th variable on the j th factor, meaning the weight that the j th factor carries in determining the i th variable, $F_1, \ldots, F_m$ are called *common factors*, and ε_i is the error (or called the specific factor) associated only with the i th variable. Here $L = (l_{ij})$ is non-random.

An illustration picture

Factor analysis (FA) model is a latent variable model.

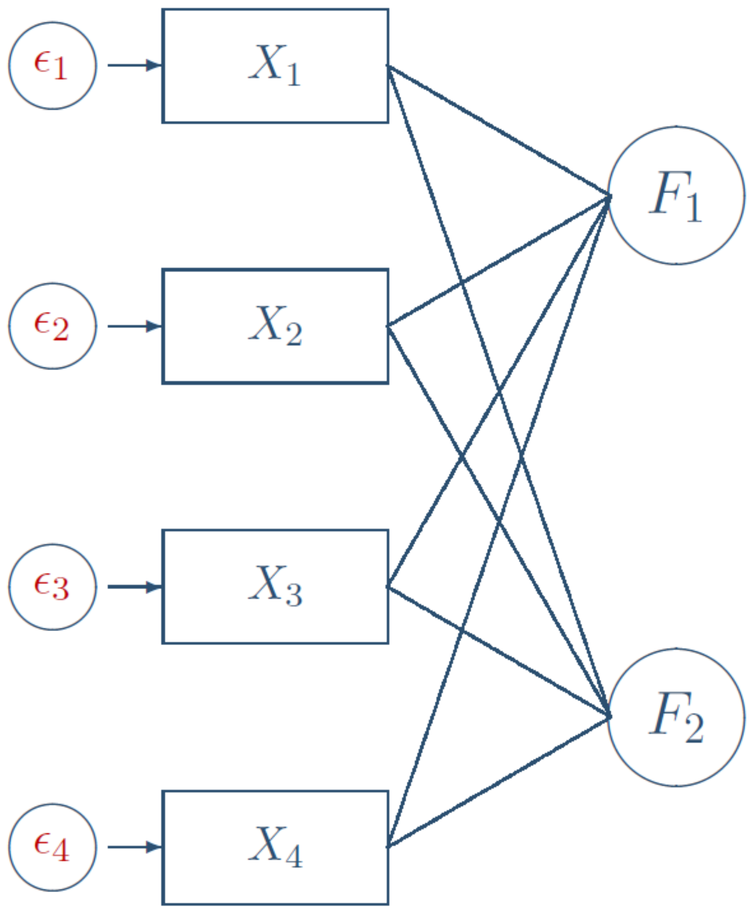
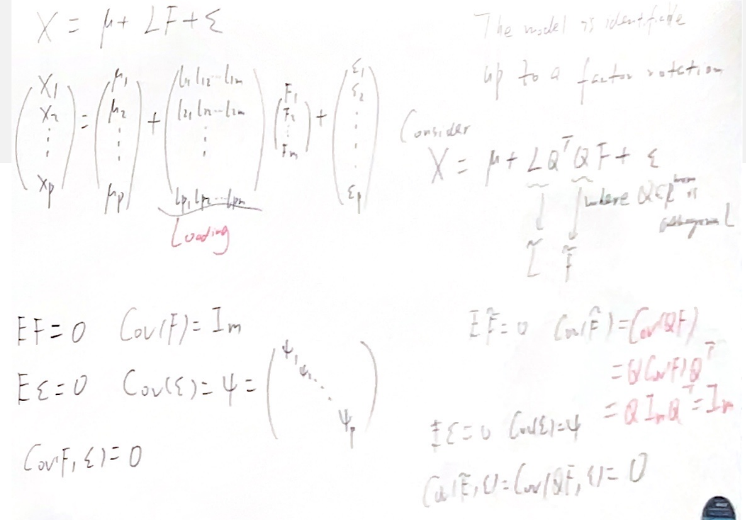


Figure: A FA model for $p = 4$ and $m = 2$

Model assumptions



- $E(\boldsymbol{F}) = \mathbf{0}$ and $Cov(\boldsymbol{F}) = \boldsymbol{I}_m$
- $E(\boldsymbol{\epsilon}) = 0$ and $Cov(\boldsymbol{\epsilon}) = \boldsymbol{\Psi} = \text{diag}\left(\psi_1, \cdots, \psi_p\right)$
- $Cov(\boldsymbol{F}, \boldsymbol{\epsilon}) = \mathbf{0}$

Remark: Thanks to the assumption $Cov(\mathbf{F}) = \mathbf{I}_m$, we have $Cov(\mathbf{QF}) = \mathbf{I}_m$ for any orthogonal matrix $\mathbf{Q}$ and the model assumptions remain unchanged if we replace $\mathbf{L} \to \mathbf{LQ}^\top$ and $\mathbf{F} \to \mathbf{QF}$. So the model is identifiable only up to a factor rotation. Thus F_i 's are called orthogonal factors and the model is also named orthogonal factor model.

Mean and Covariance structure

- Mean of $\boldsymbol{X} - \boldsymbol{\mu} = E(\boldsymbol{LF} + \boldsymbol{\epsilon}) = \boldsymbol{L}E(\boldsymbol{F}) + E(\boldsymbol{\epsilon}) = \mathbf{0}$
- Covariance matrix for $\mathbf{X}$ or equivalently $\mathbf{X} - \boldsymbol{\mu}$:

$$\begin{aligned}
\boldsymbol{\Sigma} &= E\left[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top\right] \\
&= E\left[(\boldsymbol{LF} + \boldsymbol{\epsilon})(\boldsymbol{LF} + \boldsymbol{\epsilon})^\top\right] \\
&= E\left[(\boldsymbol{LF} + \boldsymbol{\epsilon})\left(\boldsymbol{F}^\top \boldsymbol{L}^\top + \boldsymbol{\epsilon}^\top\right)\right] \\
&= \boldsymbol{L}\underbrace{E\left[\boldsymbol{FF}^\top\right]}_{\boldsymbol{I}}\boldsymbol{L}^\top + \boldsymbol{L}\underbrace{E\left[\boldsymbol{F\epsilon}^\top\right]}_{0} + \underbrace{E\left[\boldsymbol{\epsilon F}^\top\right]}_{0}\boldsymbol{L}^\top + \underbrace{E\left[\boldsymbol{\epsilon\epsilon}^\top\right]}_{\boldsymbol{\Psi}} \\
&= \boldsymbol{LL}^\top + \boldsymbol{\Psi}
\end{aligned}$$

- $Cov(\mathbf{X}, \mathbf{F}) = \mathbf{L}$

Basic quantities

Variance for X_i :

$$\sigma_{ii} = \underbrace{\sum_{k=1}^{m} \ell_{ik}^2}_{h_i^2} + \psi_i.$$

Covariance between X_i and X_j :

$$\sigma_{ij} = \sum_{k=1}^{m} \ell_{ik}\ell_{jk}.$$

Here: $\sum_{q=1}^{m} \ell_{iq}^2 = h_i^2 =$ is called i-th *communality* and ψ_i the i-th *specific variance.*

Some Remarks

- Note that the positive definite covariance can always be written as $\mathbf{A}\mathbf{A}^\top$ for some square matrix $\mathbf{A}$, but may not be written in the form $\mathbf{L}\mathbf{L}^\top + \mathbf{\Psi}$ for $m < p$.
- The model is scale invariant, meaning that if $\mathbf{X}$ follows a factor model, then so does $\mathbf{Y} = \mathbf{D}\mathbf{X}$ for any diagonal matrix $\mathbf{D}$ with rescaled $\mathbf{L}$ and $\mathbf{\Psi}$.
- Since $\mathbf{F}$ can be identified only up to orthogonal transform, we typically rotate the matrix $\mathbf{L}$ by choosing a particular version $\mathbf{L}\mathbf{Q}^\top$ for some orthogonal matrix $\mathbf{Q}$ according to an "ease of interpretation" criterion.

Table of Contents

Objective

- Given observations $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ on p generally correlated variables, factor analysis seeks to answer the question:
Does the factor model, with a small number of factors, adequately represent the data?
- We examine the covariance relationships in data. The objective of factor analysis is to find the factor loading matrix $\mathbf{L}$ and the specific variances $\boldsymbol{\Psi}$ along with selecting the number of factors m. We consider the principal component method and the maximum likelihood method.

Estimation methods

Estimation method (Factor extraction)

- Eigen-decomposition based
  - Principal component method: "principal component" solution of the factor model.
  - Iterative eigen-decompositions of $\boldsymbol{S} - \tilde{\Psi}$ (not important in this course)
- Maximum likelihood estimation: now must assume that $\boldsymbol{F}$ and $\boldsymbol{\epsilon}$ are multivariate normal. Tends to fit data better & yields scale invariance (ie., use either $\mathbf{S}$ or $\mathbf{R}$).

Estimation: Principal component method

Using spectral decomposition, we may write

$$\boldsymbol{\Sigma} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top = \sum_{j=1}^{p} \lambda_j \mathbf{u}_j \mathbf{u}_j^\top ,$$

where $\lambda_1 \geq \cdots \geq \lambda_p$ are the eigenvalues with corresponding eigenvectors $\mathbf{u}_1, \ldots, \mathbf{u}_p$. We approximate $\boldsymbol{\Sigma}$ by taking the first m terms. Define

$$\mathbf{L} = \left(\sqrt{\lambda_1}\mathbf{u}_1, \ldots, \sqrt{\lambda_m}\mathbf{u}_m\right), \quad \boldsymbol{\Psi} = \mathrm{diag}\left(\boldsymbol{\Sigma} - \mathbf{L}\mathbf{L}^\top\right).$$

Then $\boldsymbol{\Sigma} \approx \mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi}$. To apply this method to a data set the first step is to estimate $\boldsymbol{\Sigma}$ by calculating the sample covariance matrix $\mathbf{S}$, or the sample correlation matrix $\mathbf{R}$, since the factor model is scale invariant.

Principal component method

Let $\mathbf{S} = \sum_{j=1}^{p} \hat{\lambda}_j \hat{\mathbf{u}}_j \hat{\mathbf{u}}_j^\top$ be the spectral decomposition. Given m, the matrix of factor loading is

$$\hat{\mathbf{L}} = \left(\sqrt{\hat{\lambda}_1} \hat{\mathbf{u}}_1, \ldots, \sqrt{\hat{\lambda}_m} \hat{\mathbf{u}}_m \right).$$

The estimated specific variance are provided by the diagonal elements of $\mathbf{S} - \hat{\mathbf{L}}\hat{\mathbf{L}}^\top$, which is

$$\widehat{\boldsymbol{\Psi}} = \begin{bmatrix} \hat{\psi}_1 & 0 & \cdots & 0 \\ 0 & \hat{\psi}_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \hat{\psi}_p \end{bmatrix} \text{ with } \quad \hat{\psi}_i = s_{ii} - \sum_{j=1}^{m} \hat{\ell}_{ij}^2$$

Communalities are estimated as

$$\hat{h}_i^2 = \hat{\ell}_{i1}^2 + \hat{\ell}_{i2}^2 + \cdots + \hat{\ell}_{im}^2$$

The principal component factor analysis of the sample correlation matrix is obtained by starting with $\mathbf{R}$ in place of $\mathbf{S}$.

Choice of m

If m is not determined by a priori consideration, we need some criteria to estimate m. Note that the estimated loadings for a given factor do not change even if m is increased using principal component method. Consider the residual matrix

$$\mathbf{S} - \left(\hat{\mathbf{L}}\hat{\mathbf{L}}^\top + \hat{\boldsymbol{\Psi}}\right).$$

Note that this matrix has zero diagonal. It follows that

$$\left\|\mathbf{S} - \left(\hat{\mathbf{L}}\hat{\mathbf{L}}^\top + \hat{\boldsymbol{\Psi}}\right)\right\|_{HS}^2 \leq \sum_{i=m+1}^{p} \hat{\lambda}_i^2,$$

where $\|\mathbf{A}\|_{HS} = \left[\mathrm{tr}\left(\mathbf{A}\mathbf{A}^\top\right)\right]^{1/2}$ is the Hilbert-Schmidt norm. This provides a guide to choose m. Ideally, we want the RHS to be small for some small m.

Choice of m—proportion of variance explained

The contribution of the j th common factor towards the sample variance s_{ii} is $\hat{l}_{ij}^2$ since $s_{ii} = \sum_{j=1}^{m} \hat{l}_{ij}^2 + \hat{\psi}_i$. So the proportion of the total sample variance due to j-th factor is

$$\frac{\sum_{i=1}^{p} \hat{l}_{ij}^2}{\operatorname{tr} \mathbf{S}} = \frac{\sum_{i=1}^{p} (\sqrt{\hat{\lambda}_j} \hat{u}_{ji})^2}{\operatorname{tr} \mathbf{S}} = \frac{\hat{\lambda}_j}{\operatorname{tr} \mathbf{S}}$$

[1] which provide a better criterion for the choice of m. The number of common factors retained in the model is increased until a "suitable" proportion of the total sample variance has been explained.

[1] $\hat{u}_{ji}$ is the i-th coordiate of $\hat{\mathbf{u}}_j$

Stocks data example

The weekly rates of return for five stocks (JP Morgan, Citibank, Wells Fargo, Royal Dutch Shell, and ExxonMobil) listed on the New York Stock Exchange were determined for the period January 2004 through December 2005. The weekly rates of return are defined as (current week closing price-previous week closing price)/(previous week closing price), adjusted for stock splits and dividends. The observations in 103 successive weeks appear to be independently distributed.

Principal component method solution

Variable	One-factor solution		Two-factor solution		
	Estimated factor loadings F_1	Specific variances $\tilde{\psi}_i = 1 - \tilde{h}_i^2$	Estimated factor loadings F_1	Estimated factor loadings F_2	Specific variances $\tilde{\psi}_i = 1 - \tilde{h}_i^2$
1. J P Morgan	.732	.46	.732	−.437	.27
2. Citibank	.831	.31	.831	−.280	.23
3. Wells Fargo	.726	.47	.726	−.374	.33
4. Royal Dutch Shell	.605	.63	.605	.694	.15
5. ExxonMobil	.563	.68	.563	.719	.17
Cumulative proportion of total (standardized) sample variance explained	.487		.487	.769	

A look on the residual matrix

The residual matrix corresponding to the solution for $m = 2$ factors is

$$\mathbf{R} - \widehat{\mathbf{L}}\widehat{\mathbf{L}}^\top - \hat{\mathbf{\Psi}} = \begin{bmatrix} 0 & -.099 & -.185 & -.025 & .056 \\ -.099 & 0 & -.134 & .014 & -.054 \\ -.185 & -.134 & 0 & .003 & .006 \\ -.025 & .014 & .003 & 0 & -.156 \\ .056 & -.054 & .006 & -.156 & 0 \end{bmatrix}$$

The proportion of the total variance explained by the two-factor solution is appreciably larger than that for the one-factor solution. However, for $m = 2$, $\widehat{\mathbf{L}}\widehat{\mathbf{L}}^\top$ produces numbers that are, in general, larger than the sample correlations. This is particularly true for r_{13}.

Maximum likelihood method

- With principal component solution, extraction of an additional factor does not effect the values already extracted; this is not the case with the MLE.
- You get different results if you use $\mathbf{S}$ or $\mathbf{R}$; however, this is not the case for MLE (maximum likelihood estimator):
- You can get a better fit via MLE.
- Statistical tests become possible using MLE.

Maximum likelihood method

We assume $\mathbf{F}$ and $\boldsymbol{\epsilon}$ are jointly normal. The goal is to find the maximum of the likelihood subject to the constraint $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi}$. Recall that the log-likelihood of an i.i.d. sample $\mathbf{x}_1, \dots, \mathbf{x}_n$ from $N_p(\mu, \Sigma)$ is given by

$$l(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = -\frac{np}{2}\log(2\pi) - \frac{n}{2}\log|\boldsymbol{\Sigma}| - \frac{n}{2}\left[\mathrm{tr}\left(\boldsymbol{\Sigma}^{-1}\mathbf{S}_n\right) + (\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu})\right],$$

where $S_n = \frac{n-1}{n}S$. Plugging in $\boldsymbol{\mu} = \bar{\mathbf{x}}$ and $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi}$, the MLEs $\hat{\mathbf{L}}$ and $\hat{\boldsymbol{\Psi}}$ are obtained by minimizing (numerically)

$$\log\left|\mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi}\right| + \mathrm{tr}\left(\mathbf{S}_n\left(\mathbf{L}\mathbf{L}^\top + \boldsymbol{\Psi}\right)^{-1}\right).$$

Due to the lack of identifiability, we impose a further constraint to make $\mathbf{L}$ unique (up to multiplication of its columns by ± 1); that is $\mathbf{L}^\top\boldsymbol{\Psi}^{-1}\mathbf{L}$ is diagonal and the entries are distinct in decreasing order.

Stocks example: maximum likelihood method solution

	Maximum likelihood			Principal components		
	Estimated factor loadings		Specific variances	Estimated factor loadings		Specific variances
Variable	F_1	F_2	$\hat{\psi}_i = 1 - \hat{h}_i^2$	F_1	F_2	$\tilde{\psi}_i = 1 - \tilde{h}_i^2$
1. J P Morgan	.115	.755	.42	.732	−.437	.27
2. Citibank	.322	.788	.27	.831	−.280	.23
3. Wells Fargo	.182	.652	.54	.726	−.374	.33
4. Royal Dutch Shell	1.000	−.000	.00	.605	.694	.15
5. Texaco	.683	−.032	.53	.563	.719	.17
Cumulative proportion of total (standardized) sample variance explained	.323	.647		.487	.769	

Residual matrix

The residual matrix is

$$\mathbf{R} - \hat{\mathbf{L}}\hat{\mathbf{L}}^\top - \hat{\boldsymbol{\Psi}} = \begin{bmatrix} 0 & .001 & -.002 & .000 & .052 \\ .001 & 0 & .002 & .000 & -.033 \\ -.002 & .002 & 0 & .000 & .001 \\ .000 & .000 & .000 & 0 & .000 \\ .052 & -.033 & .001 & .000 & 0 \end{bmatrix}$$

The elements of $\mathbf{R} - \hat{\mathbf{L}}\hat{\mathbf{L}}^\top - \hat{\Psi}$ are much smaller than those of the residual matrix corresponding to the principal component factoring of $\mathbf{R}$ presented before. On this basis, we prefer the maximum likelihood approach.

Some remarks

- The cumulative proportion of the total sample variance explained by the factors is larger for principal component factoring than for maximum likelihood factoring. It is not surprising that this criterion typically favors principal component factoring.
- The patterns of the initial factor loadings for the maximum likelihood solution are constrained by the uniqueness condition that $\hat{\mathbf{L}}^\top \hat{\Psi}^{-1} \hat{\mathbf{L}}$ be a diagonal matrix. Therefore, useful factor patterns are often not revealed until the factors are rotated.

Factor rotation

- Since the original loadings may not be readily interpretable, it is usual practice to rotate them until a ”simpler structure” is achieved.
  Ideally, we should like to see a pattern of loadings such that each variable loads highly on a single factor and has small to moderate loadings on the remaining factors
- Mathematically, we may deliberately change the loadings $\mathbf{L} \to \mathbf{L}\mathbf{Q}^\top$ for some orthogonal matrix $\mathbf{Q}$ in order to improve the interpretability for factors $\mathbf{Q}^\top\mathbf{F}$ (rotation of factors).

Desirable structure

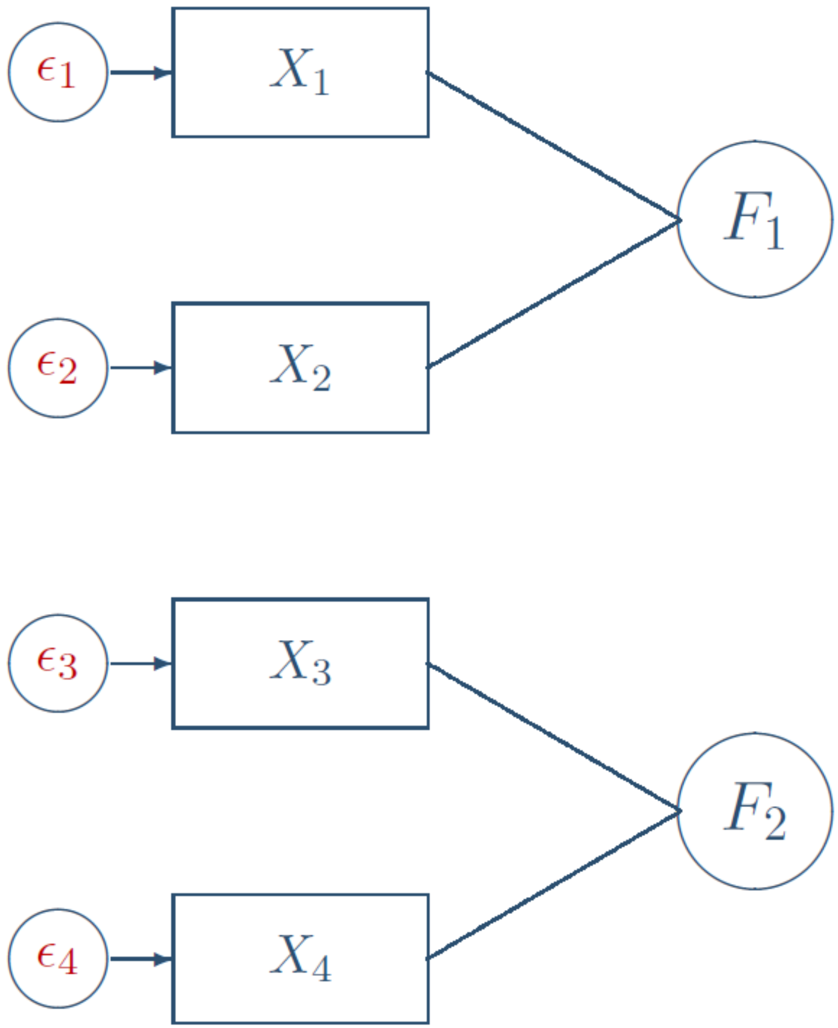


Figure: It is possible that by using proper rotation, each variable loads highly on a single factor

Varimax criterion and factor rotation

Define $\tilde{\ell}^*_{ij} = \hat{\ell}^*_{ij}/\hat{h}_i$ to be the rotated coefficients scaled by the square root of the communalities. Then the varimax procedure selects the orthogonal transformation $\mathbf{Q}$ that makes

$$V = \frac{1}{p}\sum_{j=1}^{m}\left[\sum_{i=1}^{p}\tilde{\ell}^{*4}_{ij} - \left(\sum_{i=1}^{p}\tilde{\ell}^{*2}_{ij}\right)^2 / p\right]$$

as large as possible. [2]

This has a simple interpretation. In words,

$$V \propto \sum_{j=1}^{m}\left(\begin{array}{c}\text{variance of squares of (scaled) loadings for} \\ j\text{ th factor}\end{array}\right)$$

Effectively, maximizing V corresponds to "spreading out" the squares of the loadings on each factor as much as possible.

[2] Computing algorithms exist in statistical software

Stocks example: rotated maximum likelihood estimator

Variable	Maximum likelihood estimates of factor loadings F_1	F_2	Rotated estimated factor loadings F_1^*	F_2^*	Specific variances $\hat{\psi}_i^2 = 1 - \hat{h}_i^2$
J P Morgan	.115	.755	.763	.024	.42
Citibank	.322	.788	.821	.227	.27
Wells Fargo	.182	.652	.669	.104	.54
Royal Dutch Shell	1.000	−.000	.118	.993	.00
ExxonMobil	.683	.032	.113	.675	.53
Cumulative proportion of total sample variance explained	.323	.647	.346	.647	

Figure: Results for rotated maximum likelihood estimator

Some interpretation

The rotated loadings indicate that the bank stocks (JP Morgan, Citibank, and Wells Fargo) load highly on the first factor, while the oil stocks (Royal Dutch Shell and ExxonMobil) load highly on the second factor. The two rotated factors, together, differentiate the industries. It is difficult for us to label these factors intelligently. Factor 1 represents those unique economic forces that cause bank stocks to move together. Factor 2 appears to represent economic conditions affecting oil stocks.

Table of Contents

Statistical inference: Likelihood ratio test

The MLE method has an advantage to test whether the number of factors m is adequate, or equivalently test $H_0 : \mathbf{\Sigma} = \mathbf{L}_{p\times m}\mathbf{L}^\top + \mathbf{\Psi}$ The likelihood ratio statistic is given by

$$-2\log\Lambda = -2\log\frac{\max_{H_0} e^{l(\boldsymbol{\mu},\mathbf{\Sigma})}}{\max e^{l(\boldsymbol{\mu},\mathbf{\Sigma})}} = -2\log\frac{e^{l\left(\bar{\mathbf{x}},\hat{\mathbf{L}}\hat{\mathbf{L}}^\top+\hat{\mathbf{\Psi}}\right)}}{e^{l(\bar{\mathbf{x}},\mathbf{S}_n)}} =: nF$$

where

$$F = \log\left(\frac{\left|\hat{\mathbf{L}}\hat{\mathbf{L}}^\top + \hat{\mathbf{\Psi}}\right|}{|\mathbf{S}_n|}\right) + \operatorname{tr}\left(\mathbf{S}_n\left(\hat{\mathbf{L}}\hat{\mathbf{L}}^\top + \hat{\mathbf{\Psi}}\right)^{-1}\right) - p.$$

Likelihood ratio test

Bartlett has shown that as $n \to \infty$

$$\left(n - 1 - \frac{1}{6}(2p + 5) - \frac{2}{3}m\right) F \to \chi^2_\nu$$

where $\nu = \frac{1}{2}\left[(p - m)^2 - (p + m)\right]$ is the d.f. We may perform statistical inferences based on this fact, and reject H_0 when F is larger than the critical value. (reject H_0 means the data do not fit a factor model with m common factor)

Table of Contents

Factor scores

In FA, primary interest is on the parameters in the factor model. However, the estimated values of the common factors, called factor scores, may also be required for diagnostic purposes, as well as input to a subsequent analysis (e.g., regression). Let f denote the "realized value" of the random factor F. To estimate f from observation x, we treat the parameters μ, L and Ψ as known and are given by the (PCA or MLE) estimates. The model is

$$\mathrm{x} - \mu = L\mathrm{f} + \varepsilon, \quad \varepsilon \sim (\mathbf{0}_p, \Psi)\,.$$

Regression method

With normal assumption for the orthogonal factor model, one has

$$\begin{pmatrix} X-\mu \\ F \end{pmatrix} \sim N_{p+m}\left(0, \begin{pmatrix} \Sigma = LL^\top + \Psi & L \\ L^\top & I_m \end{pmatrix}\right).$$

The conditional distribution of F given $X = \mathrm{x}$ is

$$N_m\left(L^\top \Sigma^{-1}(\mathrm{x}-\mu), I_m - L^\top \Sigma^{-1} L\right),$$

and F is then estimated, knowing x_j, by the conditional mean (after plugging in the estimated parameters)

$$\hat{\mathbf{f}}_j = \hat{L}^\top \hat{\Sigma}^{-1}\left(\mathbf{x}_j - \overline{\mathbf{x}}\right).$$

Regression method

Using the matrix identity

$$L^\top \left(LL^\top + \Psi\right)^{-1} = \left(I_m + L^\top \Psi^{-1} L\right)^{-1} L^\top \Psi^{-1},$$

we find

$$\hat{\mathbf{f}}_j = \left(I_m + \hat{L}^\top \hat{\Psi}^{-1} \hat{L}\right)^{-1} \hat{L}^\top \hat{\Psi}^{-1} \left(\mathrm{x}_j - \overline{\mathbf{x}}\right).$$

In practice, people tend to use $\hat{\Sigma} = S$, the unconstrained sample covariance matrix to reduce the effects of a (possibly) incorrect choice of m. Then the factor scores obtained by regression are

$$\hat{\mathbf{f}}_j = \hat{L}^\top S^{-1} \left(\mathbf{x}_j - \overline{\mathbf{x}}\right).$$

A concluding remark

Factor analysis is naturally appealing in behavioral and social sciences because it helps find hidden "traits" behind what we can observe. It explains complex behaviors by discovering simpler underlying patterns that we can't see directly.

Factor analysis is subjective. What matters most is whether the results make sense.